

Bayesian Time-Varying Multi-Seasonal ARMA Models

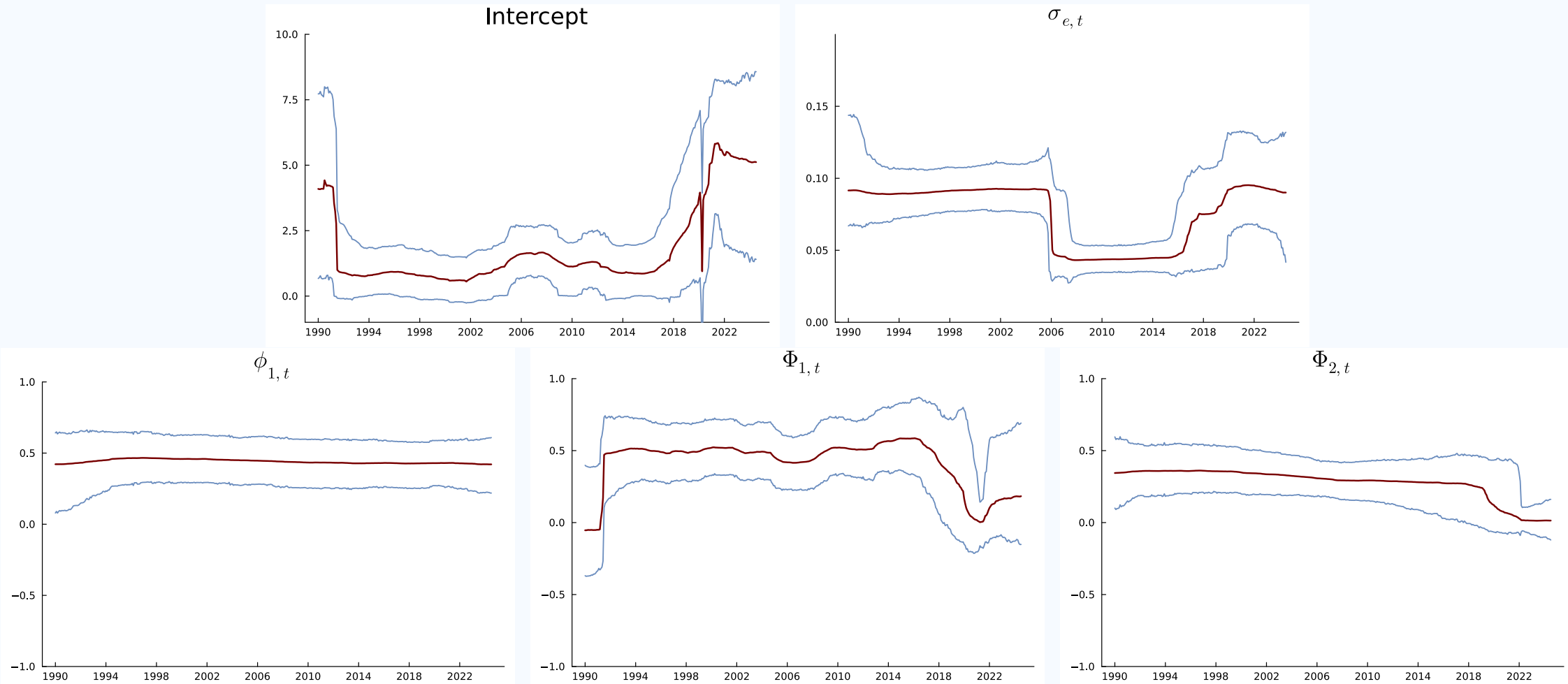
Robust, Stable and Scalable Inference for Nonstationary Time Series

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1. Why TV-SARMA Models?

Static Model (Box & Jenkins) $\implies$ TV Model (Box & Jenkins 2.0)

$$\phi(L)\Phi(L^S)y_t = \beta_0 + \psi(L)\Psi(L^S)\varepsilon_t \qquad \phi_t(L)\Phi_t(L^S)y_t = \beta_{0,t} + \psi_t(L)\Psi_t(L^S)\varepsilon_t$$



US Air Passenger Data: Posterior TV-SAR(1,2)₁₂ medians and 95% credible bands.

TV-SAR captures how major events, such as the COVID-19 pandemic, alter trend, seasonality and volatility over time. In contrast, static models assume fixed dynamics.

2. Modeling Framework and Contributions

Bayesian | TV-SARMA | Multiple Seasons | Stable & Invertible
 DSP Priors | Nonlinear Inference | Adaptive Filtering | Scalable

3. State-Space Representation

Observation: $y_t = \mathbf{x}_{0t}^\top \tilde{\phi}_t^{(AR)} + \mathbf{z}_{0t}^\top \tilde{\psi}_t^{(MA)} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$

Constraint: $\tilde{\phi}_t, \tilde{\psi}_t = \tilde{\mathbf{g}}(\theta_t)$ **(stability & invertibility)**

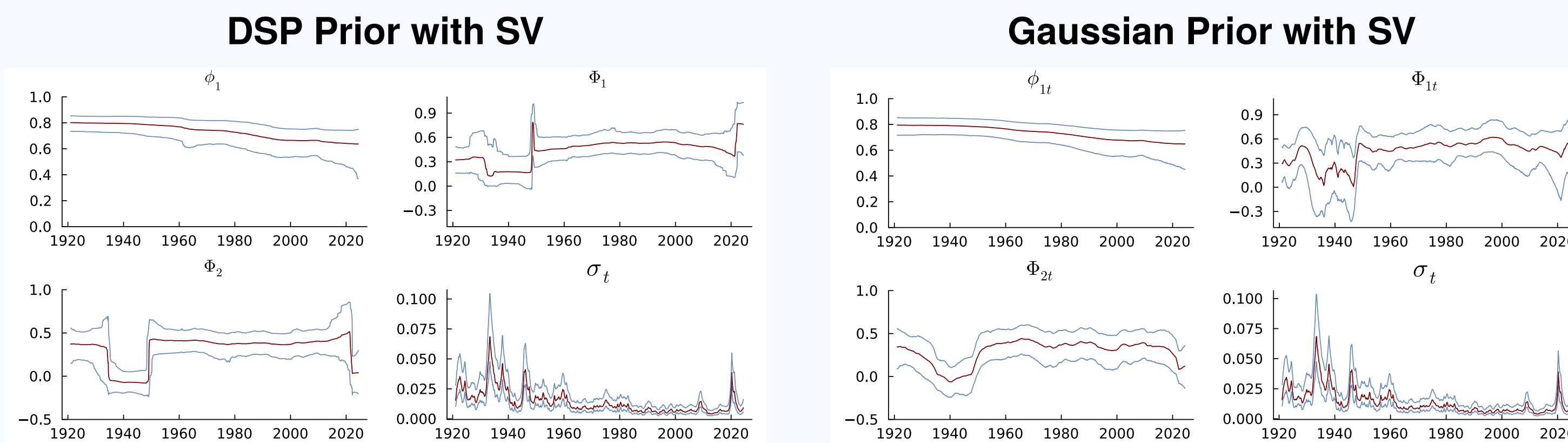
State: $\theta_t = \theta_{t-1} + \nu_t, \quad \nu_t \sim N(0, \text{Diag}\{\exp(\mathbf{h}_t)\})$

DSP: $\mathbf{h}_t = \boldsymbol{\mu} + \boldsymbol{\kappa}(\mathbf{h}_{t-1} - \boldsymbol{\mu}) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim Z(1/2, 1/2, 0, 1).$

The observation equation is the multiplicative SARMA model in nonlinear regression form. Here, $\mathbf{x}_{0t}$ and $\mathbf{z}_{0t}$ collect the nonzero lagged observations and innovations (including presample and initial values), while $\tilde{\phi}_t$ and $\tilde{\psi}_t$ contain the corresponding nonzero AR and MA coefficients, including interaction terms.

The formulation naturally accommodates multiple seasonal periods.

4. Dynamic Shrinkage vs. Gaussian Regularization

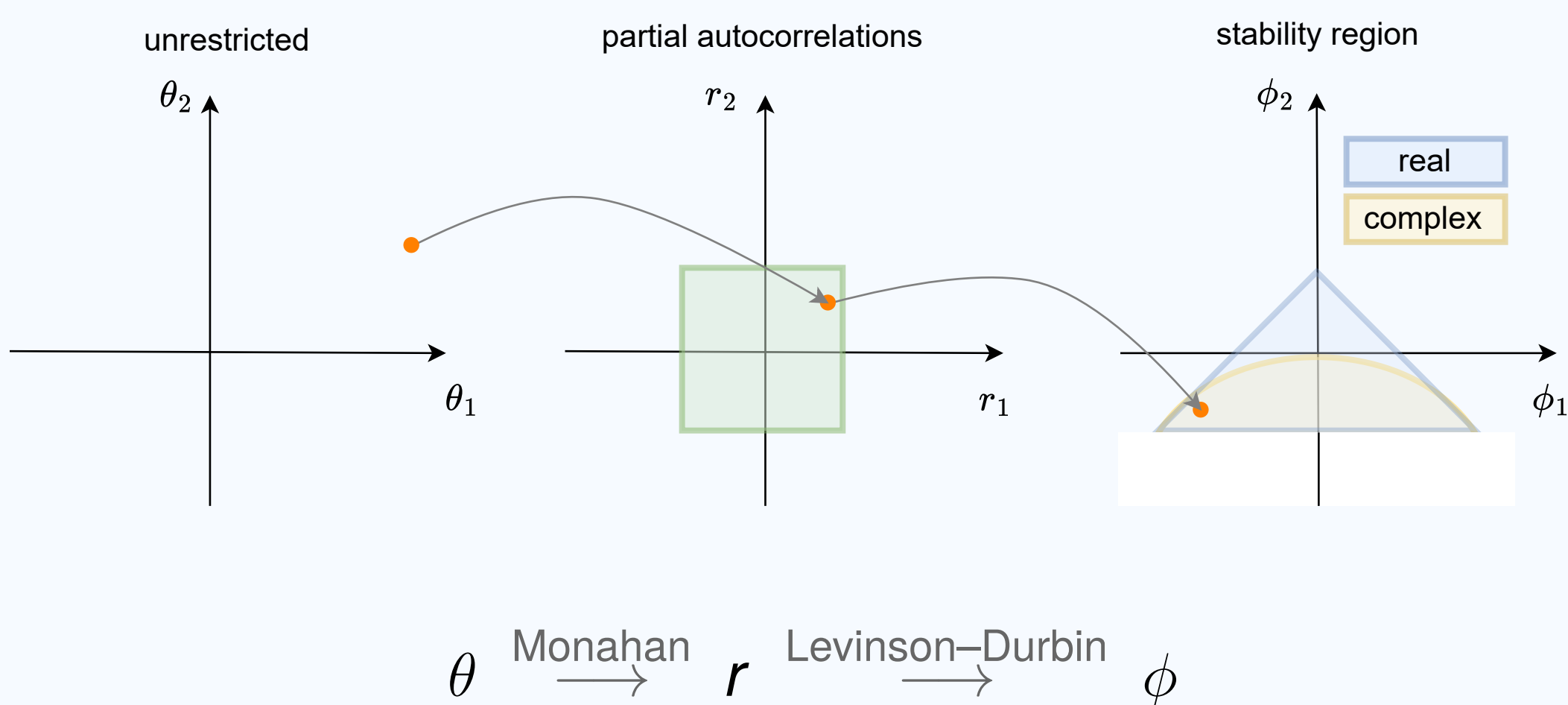


US Industry Production Index Data: Posterior TV-SAR(1,2)₁₂ medians and 95% credible bands.

DSP prior (left): adaptive shrinkage, smoother trajectories, and abrupt-change detection.

Gaussian prior (right): less regularization, more variable trajectories.

5. Stability Parametrization and Uniform Prior

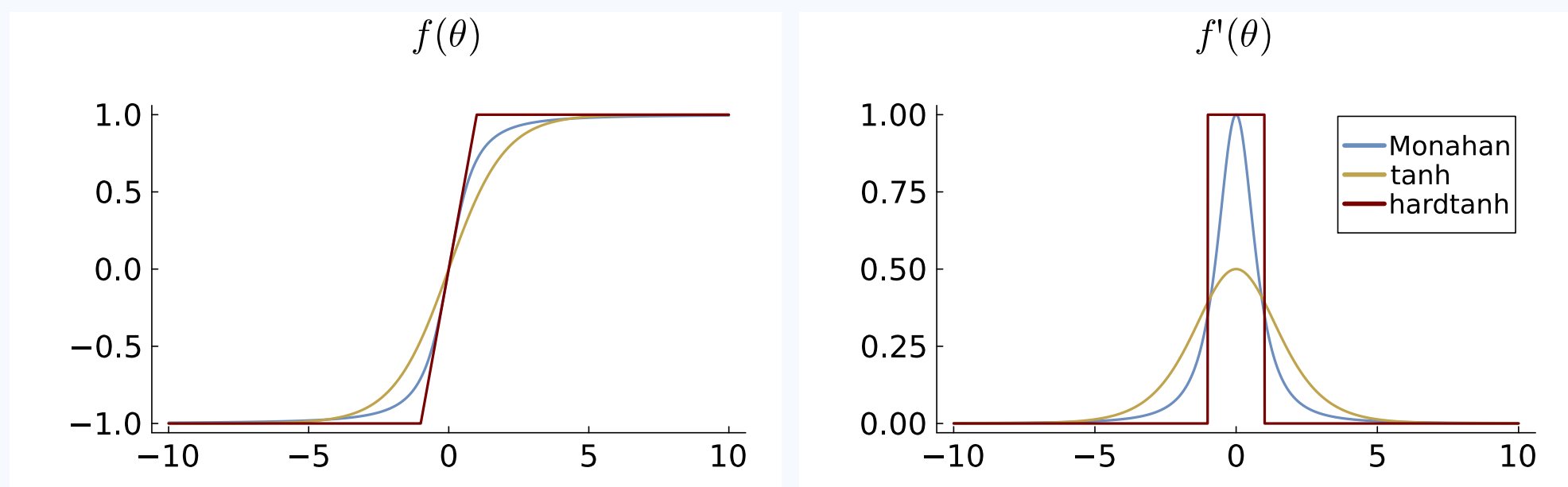


Uniform prior over the stable AR region with Monahan (1984)

Same idea applies to ensure invertibility!

6. Robust Stability Parametrization

Monahan (1984) $\implies \tanh \implies$ **hardtanh**

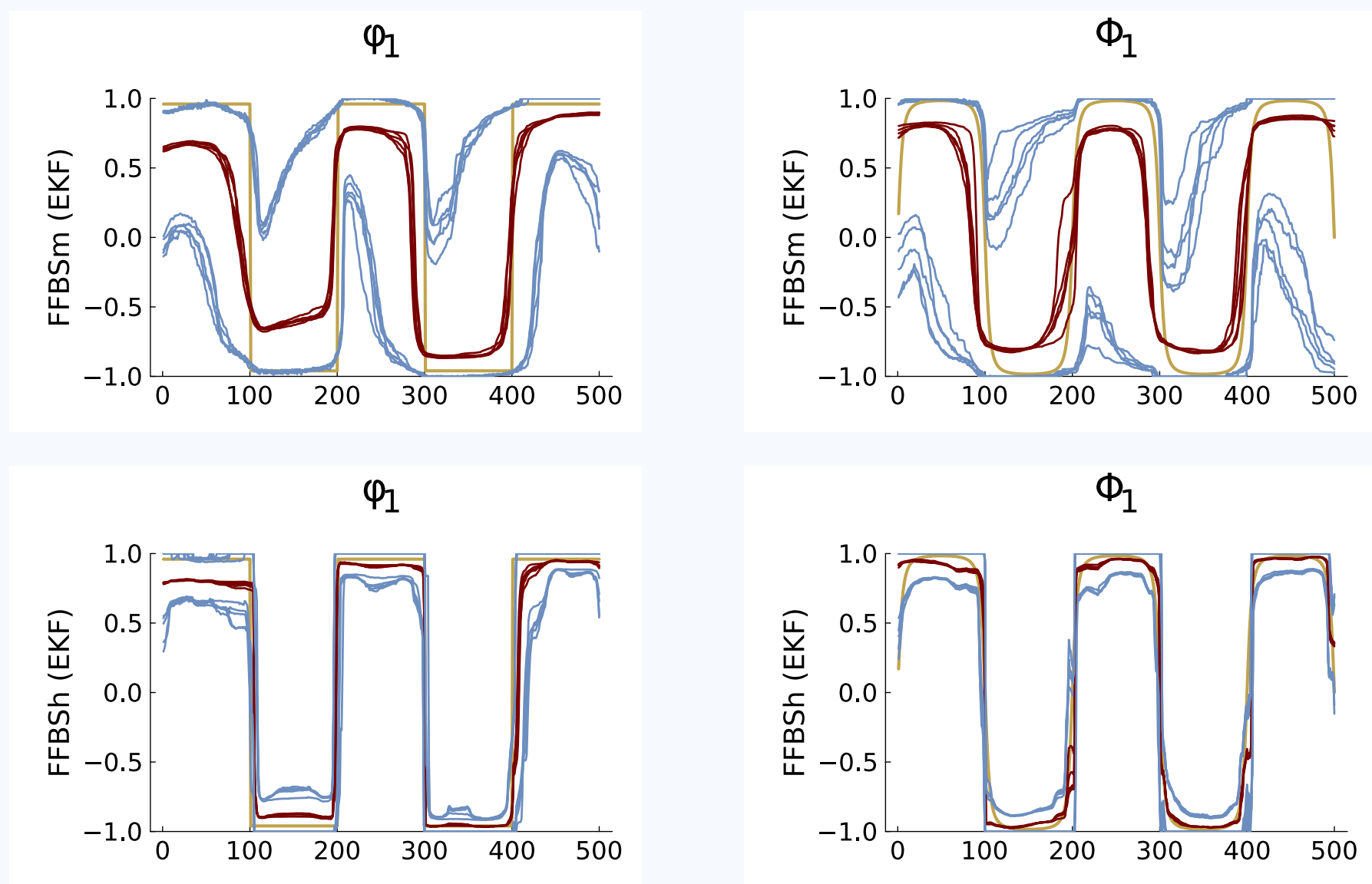


$$\frac{\theta}{\sqrt{1+\theta^2}} \longrightarrow \tanh(\theta/2) \longrightarrow \min(1, \max(-1, \theta))$$

(vanishing gradients) (vanishing gradients) **(linear interior)**

7. Near the Stability Boundary

TV-SAR(1,1)₁₂ | Extreme Boundary Dynamics | Simulated Data



Posterior medians and 95% credible bands from five independent runs under **Monahan** (first row) and **hardtanh** (second row) parametrizations.

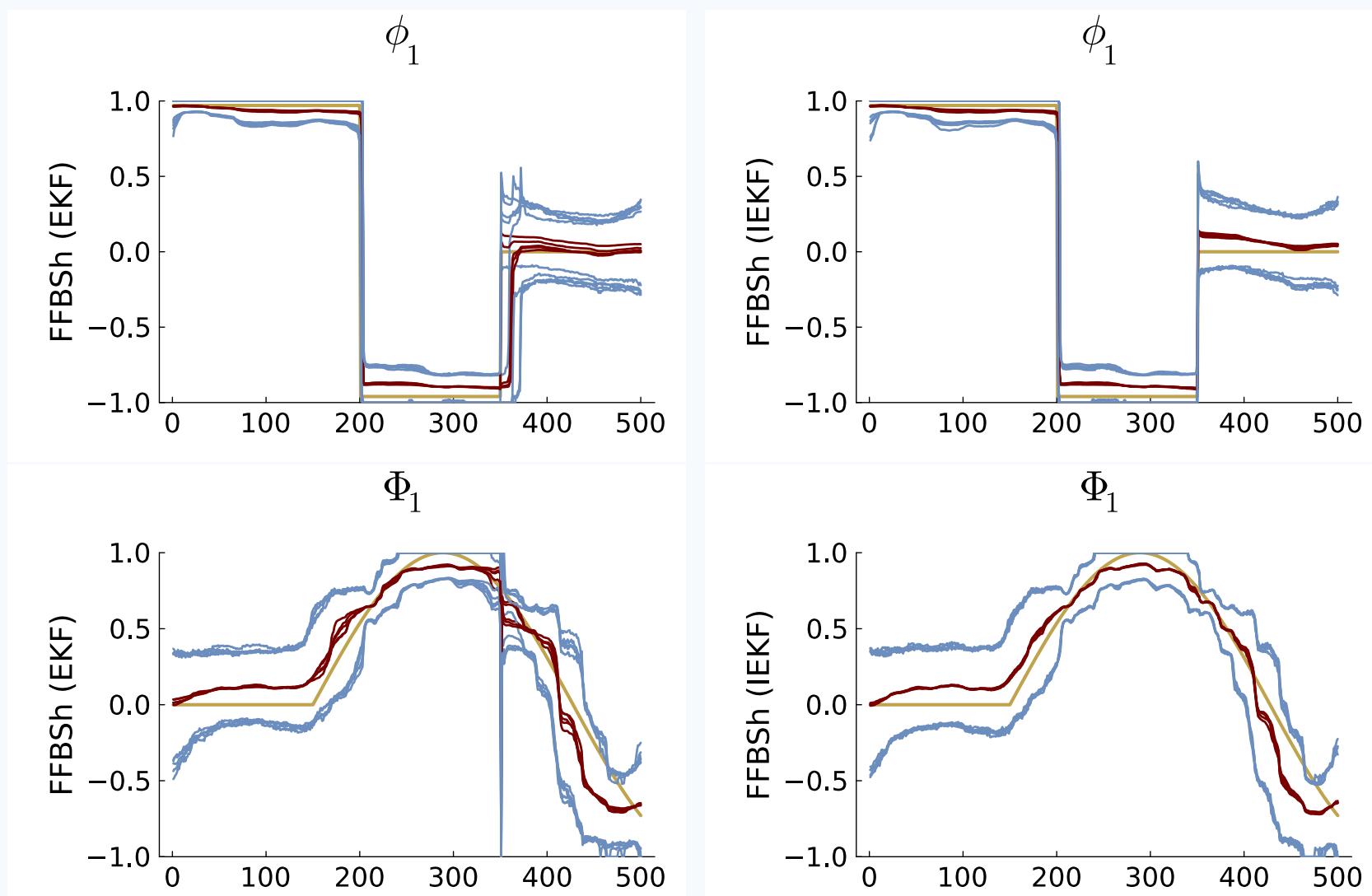
Monahan (first row): unstable filtering and inflated credible bands.

Hardtanh + constrained filtering (second row): robust, numerically stable inference.

8. Robust Adaptive Inference

Hardtanh | Constrained Filtering | Adaptive Inference

FFBS+EKF (Fast) $\xrightarrow{\text{NIS trigger}}$ **FFBS+IEKF** (Accurate)



Simulated TV-SAR(1,1)₁₂: posterior medians and 95% credible bands from five independent runs under **FFBS+EKF** (left column) and **FFBS+IEKF** (right column).

Adaptive inference: fast when possible, accurate when necessary!

9. Gibbs sampling scheme

States $\implies$ DSP $\implies$ Presample $\implies$ Errors $\implies$ Volatility
 (FFBS+(I)EKF) (Kowal, 2017) (Closed-form) (Closed-form) (SV+DSP)

10. Scaling to Multiple Seasonalities

Grouped Computation $\implies$ Prior-Preserving

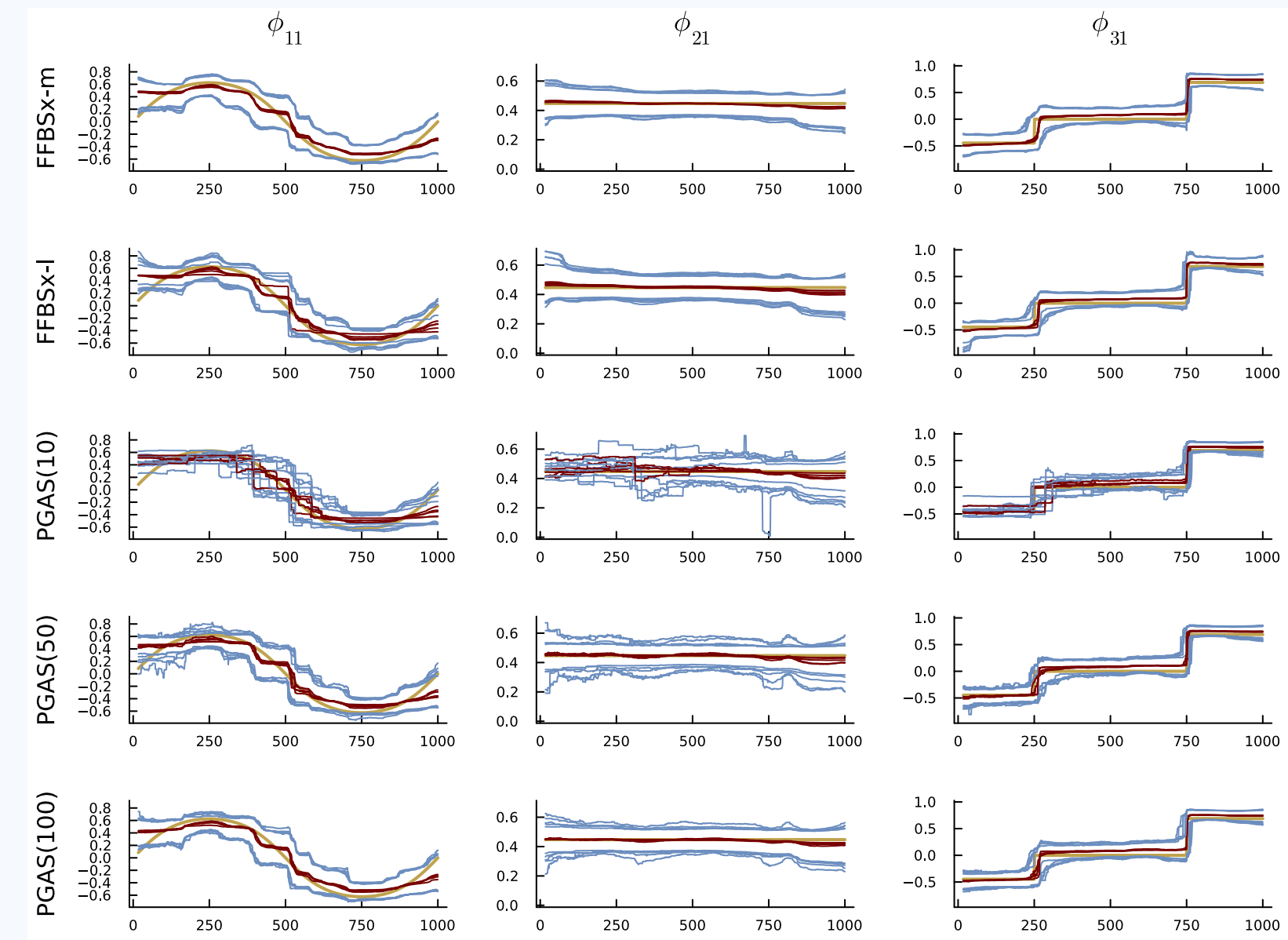
(Faster inference) (Group-adjusted innovations)

Grouping $\implies$ Sequential EKF $\implies$ **Scalable Inference**

(Fewer state updates) (Lower filtering cost) (Long high-frequency series)

Grouping is a computational device, not a modelling assumption.

11. Simulations



TV-SAR(1,1,1)_(4,12): Posterior medians (red) and 95% credible bands (blue) from five independent runs.



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